

About Simulation Object Standard - (SOS)

Defining What the Simulation Actually Represents

Actually Represents

Every simulation begins with an object.

That object may be a machine, system, process, environment, behavior, population, mission, infrastructure, autonomous system, AI-enabled system, or another target intended to be represented through simulation.

Before anything can be simulated, one question must be answered:

What exactly are we simulating?

Simulation Object Standard (SOS) establishes the governance conditions required to answer that question consistently.

SOS is the first Parent Standard of SIMULOS® — Simulation Governance Architecture.

It governs the Simulation Object before simulation purpose, representation, assumptions, models, scenarios, execution, outputs, or reality correlation begin.

Why SOS Exists

Simulation can produce highly precise outputs while still representing the wrong object.

This may occur when: the object is poorly defined, the wrong version is simulated, configurations are mixed, important components are excluded, system boundaries are unclear, a proxy is treated as the real target, or the object changes while the simulation continues to rely on an older definition.

SOS prevents these problems by establishing a stable object foundation for the complete simulation lifecycle.

What Is a Simulation Object?

A Simulation Object is the governed target that a simulation intends to represent.

It defines:

what belongs to the simulation target,

what remains outside it,

which version applies,

which configuration applies,

which state applies,

and:

which object the resulting simulation outputs actually concern.

A Simulation Object may be simple or highly complex.

It may represent one component or an entire system-of-systems.

The key requirement is that its identity and boundaries remain explicit.

Object Before Representation

SOS separates the real or intended simulation target from the mechanisms used to simulate it.

Simulation Object ≠ Simulation Model

Simulation Object ≠ Simulation Representation

Simulation Object ≠ Simulation Environment

The Simulation Object is what is being simulated.

The model, representation, and environment are mechanisms through which the object is later represented and examined.

This distinction prevents simulation technology from redefining the object merely because a particular model is easier to construct.

Object Identity and Boundary

A governed Simulation Object must be sufficiently distinguishable from materially different objects.

Depending on the context, this may require identification of: system identity, version, configuration, relevant components, operational state, dependencies, temporal condition, and object boundaries.

The object boundary establishes what the simulation claim can legitimately refer to.

If only part of a system is included, the resulting simulation must not be represented as though the entire system was simulated.

Versions, Configurations, and Change

Simulation relevance may depend on the exact object version and configuration.

A simulation of:

Version A

does not automatically apply to:

Version B.

A simulation of:

Configuration X

does not automatically apply to:

Configuration Y.

SOS therefore keeps simulation results connected to the object state they actually represent.

When the object materially changes, that change must remain visible to downstream SIMULOS® governance.

Complex Simulation Objects

SIMULOS® may govern simulations involving: autonomous vehicles, aircraft, spacecraft, robot fleets, AI systems, industrial facilities, digital twins, infrastructure networks, organizations, populations, environmental systems, and complete system-of-systems environments.

SOS allows complex objects to be decomposed into identifiable components while preserving their relationship to the overall Simulation Object.

This makes it possible to understand whether a simulation conclusion applies to:

a component,

a subsystem,

a complete system,

or:

an interconnected operational environment.

Preventing Object Inflation

One of the central functions of SOS is to prevent a narrow simulation from becoming a broader claim than the object definition supports.

SOS therefore protects against:

component → whole-system inflation

single-version → product-family inflation

single-configuration → all-configurations inflation

and:

proxy-object → target-object substitution

The simulation claim must remain anchored to the object actually simulated.

Governance Flow

SOS progresses through:

Candidate Simulation Target → Simulation Object Identification → Object
Boundary → Object Structure → Object Version → Object Configuration → Object
State → Object Documentation → Governed Simulation Object

Once the Simulation Object is sufficiently established, SIMULOS® can proceed
to the next question:

Why is this object being simulated, and what must the simulation cover?

Relationship to SIMULOS®

SOS establishes the first governance foundation of the complete SIMULOS®
lifecycle.

The architecture then progresses into:

Simulation Purpose & Scope → Reality Representation → Assumptions → Model &
Environment → Inputs & Data → Scenario Design → Execution → Output Governance
→ Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary
→ Lifecycle & Resimulation

Every downstream stage depends upon the object established by SOS.

Relationship to Other OOF® Architectures

SOS governs specifically the Simulation Object.

ORA™ governs operational reality.

OBIDENTITY® may govern broader identity, origin, ownership, provenance, and
authority.

INTEGROS® governs integrity.

VALIDOS® governs validation.

ADIT® governs continuous audit and evidence.

These architectures may interact with SOS, but they do not replace its
simulation-specific function.

Why It Matters

A sophisticated simulation cannot compensate for an incorrectly defined object.

If the object is wrong, outdated, incomplete, or ambiguous, downstream simulation precision may create only precise answers about the wrong target.

SOS therefore establishes a simple but fundamental rule:

Define the object before simulating the object.

Foundational Principle

A simulation cannot legitimately claim to represent an object more precisely than that object has been identified, bounded, versioned, configured, and documented.

Canonical Closing Statement

Simulation Object Standard (SOS) establishes the object foundation of SIMULOS® by ensuring that every simulation begins with a clearly identified, bounded, versioned, configured, state-aware, and traceable Simulation Object. By separating the simulation target from its model, representation, and environment, SOS ensures that every downstream simulation activity remains connected to the exact object it claims to represent.

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